

Assignment 4: Object-Oriented Programming in Python

Question 1: Library Management System

Problem Statement:

You are required to create a simple Library Management System using OOP concepts in Python. The system should be able to perform the following operations:

- **Add a New Book:** Allow the librarian to add a new book to the library with details such as the title, author, and ISBN number.
- **Issue a Book:** Allow the librarian to issue a book to a student. The system should check if the book is available before issuing it.
- **Return a Book:** Allow the librarian to accept the return of a book and update its availability status.
- **Check Book Availability:** Allow the librarian to check if a specific book is available or issued.

Requirements:

1. Define a `Book` class with attributes like `title`, `author`, `isbn`, and `is_issued` (to track the availability of the book).
2. Implement methods in the `Book` class to:
 - Mark the book as issued.
 - Mark the book as returned.
 - Display book details.
3. Define a `Library` class with an attribute `books` that stores a list of `Book` objects.
4. Implement methods in the `Library` class to:
 - Add a new book to the library.
 - Issue a book to a student.
 - Return a book.
 - Check the availability of a book.

Question 2: Fraction Addition using OOP

Problem Statement:

Create a Python program that allows users to add two fractions using Object-Oriented Programming concepts. The program should define a `Fraction` class to represent fractions and implement methods to add two fractions and simplify the result.

Requirements:

1. Define a **Fraction** class with attributes **numerator** and **denominator**.
2. Implement methods in the **Fraction** class to:
 - **Add two fractions:** Add the current fraction with another fraction and return a new **Fraction** object representing the sum.
 - **Simplify the fraction:** Simplify the resulting fraction to its lowest terms using the Greatest Common Divisor (GCD).
 - **Display the fraction:** Provide a method to display the fraction in the form "numerator/denominator".
3. Handle cases where the denominators are different by finding the Least Common Denominator (LCD) before performing the addition.